MOBILE_TESTING_GUIDE

NetworkPeer Mobile Testing Guide

> Complete guide for testing the Android and iOS apps on Mac M1

Prerequisites

Mac M1 Environment Setup

```
# Java 17 (ARM64) - required for Android build
brew install openjdk@17
export JAVA_HOME="/opt/homebrew/Cellar/openjdk@17/17.0.20.1/libexec/openjdk.jdk/Contents/Home"
export PATH="$JAVA_HOME/bin:$PATH"

# Verify
arch -arm64 java -version
# Should show: openjdk version "17.0.x" ... aarch64

# Android SDK
mkdir -p ~/Library/Android/sdk
export ANDROID_HOME="$HOME/Library/Android/sdk"
export PATH="$ANDROID_HOME/platform-tools:$ANDROID_HOME/cmdline-
        tools/latest/bin:$PATH"

# Install SDK components (if not already installed)
sdkmanager "platforms;android-36" "build-tools;35.0.0" "platform-
        tools"
```

Option A: Physical Android Device (Recommended)

1. Enable Developer Options on Android

1. **Settings** → **About Phone** → **Software Information**
2. Tap **Build Number** 7 times → "You are now a developer!"

3. **Settings** → **Developer Options** → Enable **USB Debugging**

2. Connect to Mac

```
# Connect via USB-C
# Allow USB debugging on phone prompt

# Verify connection
adb devices
# Should show: <serial>    device
```

3. Build & Install

```
export JAVA_HOME="/opt/homebrew/Cellar/openjdk@17/17.0.20.1/libexec/openjdk.jdk/Contents/Home"
export ANDROID_HOME="$HOME/Library/Android/sdk"

cd apps/android
./gradlew installDevelopmentDebug --no-daemon

# Or build APK first, then install:
./gradlew assembleDevelopmentDebug --no-daemon
adb install -r app/build/outputs/apk/development/debug/app-
        development-debug.apk
```

4. Launch & Test

```
# Launch app
adb shell monkey -p com.networkpeer.mobile.dev -c
        android.intent.category.LAUNCHER 1

# View logs
adb logcat -s "NetworkPeer:*"

# Or filter by tag
adb logcat -s "NetworkPeerAPI" "NetworkPeerUI"
        "NetworkPeerCamera"
```

5. Screen Mirroring (for demo)

```
# Install scrcpy
brew install scrcpy

# Mirror phone screen to Mac
scrcpy --stay-awake --turn-screen-on --power-off-on-close=false
```

Option B: Android Emulator (No Physical Device)

1. Create ARM64 Emulator

```
# Open Android Studio → More Actions → Virtual Device Manager
# Or use command line:
avdmanager create avd -n "pixel7_arm64" -k "system-
        images;android-36;default;arm64-v8a" -d pixel_7
```

2. Launch Emulator

```
# With GPU acceleration for M1
emulator -avd pixel7_arm64 -gpu host -no-audio -no-boot-anim

# Wait for boot, then verify
adb devices
```

3. Build & Install

```
cd apps/android
./gradlew installDevelopmentDebug --no-daemon
```

4. Simulate Camera & Location

- **Camera**: Emulator sidebar → Camera → “Virtual Scene” or “Webcam0”
- **Location**: Emulator sidebar → Location → Set GPS coordinates

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Option C: iOS Testing (Mac M1 Only)

iOS Simulator

```
cd apps/ios
open NetworkPeer.xcworkspace

# In Xcode:
# 1. Select "iPhone 15" or "iPhone 16" simulator
# 2. Press Cmd+R to build and run
```

Physical iPhone

```
# 1. Connect iPhone via USB-C/Lightning
# 2. Trust computer on iPhone
# 3. In Xcode: Window → Devices and Simulators → select iPhone
# 4. Set Signing Team: Project → Target → Signing & Capabilities
        → Team: Your Apple ID
```

```
# 5. Press Cmd+R to deploy
# 4. On iPhone: Settings → General → VPN & Device Management →
        Trust Developer Profile
```

API Configuration for Local Testing

The app points to production API by default. To test against local backend:

1. Start Local Backend

```
cd NetworkPeer-main
npm install
npm run dev  # Runs on http://localhost:3000
```

2. Configure Android for Local API

```
# Edit apps/android/networkpeer.development.local.properties
API_BASE_URL=http://10.0.2.2:3000/api/v1/
STRIPE_PUBLISHABLE_KEY=pk_test_your_key
REALTIME_URL=http://10.0.2.2:3000
REALTIME_ORIGIN=
```

Note: `10.0.2.2` is the Android emulator's alias for the host machine's localhost.

3. Rebuild & Test

```
cd apps/android
./gradlew assembleDevelopmentDebug --no-daemon
adb install -r app/build/outputs/apk/development/debug/app-
        development-debug.apk
```

Test Scenarios

1. Post a Job Flow

1. Open app → Login/Register (phone + OTP)
2. Navigate to "Post a Job"
3. Complete 6-step wizard:
    - **Job Info**: Title, description, location
    - **Task Type**: Collection / Correction
    - **Media**: Photo/Video requirements
    - **Capacity**: Single / Capped / Unlimited
    - **Escrow**: Per-unit pricing

- **Review**: Preview & submit

2. Worker Acceptance

1. Switch to Worker role
2. Browse available jobs
3. Accept a job → View assignment
4. Capture evidence using in-app camera
5. Submit for review

3. Review Flow

1. Switch to Client/Correctionist role
2. Open Review Queue
3. Open SubmissionReviewPane:
   - Zoom/pan image viewer
   - OCR results panel
   - Quality metrics (edge coverage, sharpness, exposure)
4. Approve / Redo / Reject

4. Offline & Sync

1. Enable airplane mode
2. Complete job steps
3. Re-enable network
4. Verify sync completes

Troubleshooting

Issue	Fix
`adb devices` empty	Try different USB cable, enable USB debugging, `adb kill-server && adb start-server`
Gradle build fails	`./gradlew clean && ./gradlew assembleDevelopmentDebug`
Emulator won't start	Enable KVM: `ls -la /dev/kvm`, use `-gpu host` flag
App crashes on launch	Check `adb logcat -s "NetworkPeer:*"`

Issue	Fix
API calls fail	Verify API_BASE_URL, check network security config
Camera permission denied	Settings → Apps → NetworkPeer → Permissions → Camera
iOS build fails	`cd apps/ios && pod install && xcodebuild clean`

Quick Reference Commands

```
# Build everything
export JAVA_HOME="/opt/homebrew/Cellar/openjdk@17/17.0.20.1/libexec/openjdk.jdk/Contents/Home"
export ANDROID_HOME="$HOME/Library/Android/sdk"

# Android
cd apps/android && ./gradlew assembleDevelopmentDebug --no-daemon

# Install on device
adb install -r app/build/outputs/apk/development/debug/app-
        development-debug.apk

# Logs
adb logcat -s "NetworkPeer:*" -v time

# Clear logs
adb logcat -c

# Screenshot
adb exec-out screencap -p > screenshot.png

# Screen record
adb exec-out screenrecord --output-format=h264 - > recording.mp4
```